# Discrete Homework 5

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## Problem 1

A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them.

**(a) How many balls must she select to be sure of having at least three balls of the same color?**

*Answer: 5.*

*Justification.* There are two colors. In the worst case she alternates colors to delay a triple as long as possible: after 4 draws she could have 2 red and 2 blue, which has no color appearing three times. The next draw (the 5th) must create a third ball of one of the two colors by the pigeonhole principle. Therefore 5 draws are sufficient and 4 are not.

**(b) How many balls must she select to be sure of having at least three blue balls?**

*Answer: 13.*

*Justification.* To delay getting three blues as long as possible, she could first draw all 10 red balls before drawing any blue balls. After that, she might draw at most 2 blue balls without reaching three. Hence with 12 draws (10 red + 2 blue) she could still have fewer than three blue balls. The next draw (the 13th) must then be blue, giving at least three blue balls. Thus 13 draws are sufficient, and 12 are not.

## Problem 2

There are 51 houses on a street. Each house has an address between 1000 and 1099, inclusive. Show that at least two houses have addresses that are consecutive integers.

**Solution.** Partition the 100 available addresses into the 50 disjoint pairs

$$(1000, 1001),\ (1002, 1003),\ \ldots,\ (1098, 1099).$$

These pairs are the "boxes." Placing the 51 houses into these 50 boxes by their addresses, the pigeonhole principle implies some box receives at least two houses. The two addresses in any box differ by exactly 1, so there exist at least two houses with consecutive addresses. □

## Problem 3. Use induction to prove each statement.

### (a) For $n \in \mathbb{N}$ with $n \geq 2$, we have $n! < n^n$.

**Base case:** $n = 2$: $2! = 2 < 4 = 2^2$.
**Inductive step:** Assume $k! < k^k$ for some $k \geq 2$. Then

$$(k+1)! = (k+1)\,k! < (k+1)\,k^k \leq (k+1)\,(k+1)^k = (k+1)^{k+1},$$

since $k^k \leq (k+1)^k$. Hence $n! < n^n$ for all $n \geq 2$. □

**(b) For $n \in \mathbb{N}$ with $n \geq 1$, $\displaystyle\sum_{i=1}^{n} i^3 = \Big(\sum_{i=1}^{n} i\Big)^2$.**

**Base case:** $n = 1$: $1^3 = 1 = (1)^2$.
**Inductive step:** Assume $\sum_{i=1}^{k} i^3 = \big(\sum_{i=1}^{k} i\big)^2 = \big(\frac{k(k+1)}{2}\big)^2$. Then

$$\sum_{i=1}^{k+1} i^3 = \left(\sum_{i=1}^{k} i^3\right) + (k+1)^3 = \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 = \left(\frac{(k+1)(k+2)}{2}\right)^2 = \left(\sum_{i=1}^{k+1} i\right)^2.$$

Thus the identity holds for all $n \geq 1$. $\square$

**(c) For all $n \in \mathbb{N}$, $5 \mid \big(2^{2n+1} + 3^{2n+1}\big)$.**

**Base case:** $n = 0$: $2^1 + 3^1 = 5$, divisible by 5.
**Inductive step:** Assume $5 \mid \big(2^{2k+1} + 3^{2k+1}\big)$. Then

$$2^{2(k+1)+1} + 3^{2(k+1)+1} = 2^{2k+3} + 3^{2k+3} = 4 \cdot 2^{2k+1} + 9 \cdot 3^{2k+1}.$$

Working modulo 5, note $4 \equiv -1$ and $9 \equiv -1$. Hence

$$4 \cdot 2^{2k+1} + 9 \cdot 3^{2k+1} \equiv -\big(2^{2k+1} + 3^{2k+1}\big) \equiv 0 \pmod 5,$$

so 5 divides $2^{2(k+1)+1} + 3^{2(k+1)+1}$. By induction, the claim holds for all $n$. $\square$

**(d) Let $F_k$ be the $k$-th Fibonacci number. For every $k \in \mathbb{N}$, $F_{3k}$ is even.**

**Base case:** $k = 1$: $F_3 = 2$ is even.
**Inductive step:** Assume for some $k \geq 1$ that

$$F_{3k} \text{ is even}, \qquad F_{3k+1} \text{ is odd}, \qquad F_{3k+2} \text{ is odd}.$$

Then using $F_{n+2} = F_{n+1} + F_n$,

$$F_{3k+3} = F_{3k+2} + F_{3k+1} = \text{odd} + \text{odd} = \text{even}.$$

Moreover,

$$F_{3k+4} = F_{3k+3} + F_{3k+2} = \text{even} + \text{odd} = \text{odd}, \quad F_{3k+5} = F_{3k+4} + F_{3k+3} = \text{odd} + \text{even} = \text{odd}.$$

Thus the parity pattern (even, odd, odd) repeats every three indices, so $F_{3(k+1)}$ is even. By induction, $F_{3k}$ is even for all $k \in \mathbb{N}$. $\square$

## Problem 4

The Lucas numbers satisfy $L_n = L_{n-1} + L_{n-2}$ with $L_0 = 2$ and $L_1 = 1$.

**(a) Show that $L_n = F_{n-1} + F_{n+1}$ for $n \geq 2$, where $F_n$ is the $n$-th Fibonacci number.**

**Proof by induction on $n$.** For $n = 2$: $L_2 = 3$ and $F_1 + F_3 = 1 + 2 = 3$. Assume for some $k \geq 2$ that $L_k = F_{k-1} + F_{k+1}$ and $L_{k-1} = F_{k-2} + F_k$. Then

$$L_{k+1} = L_k + L_{k-1} = (F_{k-1} + F_{k+1}) + (F_{k-2} + F_k) = (F_{k-2} + F_{k-1}) + (F_k + F_{k+1}) = F_k + F_{k+2},$$

which is the desired form with $n = k + 1$. Hence $L_n = F_{n-1} + F_{n+1}$ for all $n \geq 2$. $\square$

### (b) Find an explicit formula for the Lucas numbers.

The characteristic equation of $L_n = L_{n-1} + L_{n-2}$ is $x^2 = x + 1$ with roots $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Solving $L_n = A\varphi^n + B\psi^n$ using $L_0 = 2$ and $L_1 = 1$ gives $A = B = 1$. Thus

$$\boxed{L_n = \varphi^n + \psi^n}.$$

## Problem 5. Find the solution to each recurrence.

### (a) $a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3}$, with $a_0 = 3$, $a_1 = 6$, $a_2 = 0$.

Characteristic: $r^3 - 2r^2 - r + 2 = (r-2)(r-1)(r+1) = 0$. So $a_n = A \cdot 2^n + B \cdot 1^n + C(-1)^n$. Solving for the constants from the initial values yields $A = -1$, $B = 6$, $C = -2$. Therefore

$$\boxed{a_n = 6 - 2(-1)^n - 2^n}.$$

### (b) $a_n = 2a_{n-1} + 2^n$, with $a_0 = 2$.

Solve $a_n - 2a_{n-1} = 2^n$. A particular solution is $p_n = n2^n$; the homogeneous part is $c\,2^n$. Using $a_0 = 2$ gives $c = 2$. Hence

$$\boxed{a_n = (n+2)2^n}.$$

### (c) $a_n = 2a_{n-1} + 2n^2$, with $a_1 = 4$.

Try $p_n = an^2 + bn + c$. Plugging into $a_n - 2a_{n-1} = 2n^2$ gives $a = -2$, $b = -8$, $c = -12$. With homogeneous part $c_0\,2^n$ and $a_1 = 4$ we get $c_0 = 13$. Thus

$$\boxed{a_n = 13 \cdot 2^n - 2n^2 - 8n - 12}.$$

### (d) $a_n = 4a_{n-1} - 3a_{n-2} + 2^n + n + 3$, with $a_0 = 1$, $a_1 = 4$.

Homogeneous: $r^2 - 4r + 3 = (r-1)(r-3)$; so $A \cdot 1^n + B \cdot 3^n$. For $2^n$ try $C\,2^n$; for $n+3$ try a quadratic $qn^2 + pn + q_0$. Solving $a_n - 4a_{n-1} + 3a_{n-2} = 2^n + n + 3$ yields $C = -4$, $q = -\frac{1}{4}$, $p = -\frac{5}{2}$. Using the initial conditions gives $A = \frac{1}{8}$, $B = \frac{39}{8}$. Therefore

$$\boxed{a_n = \frac{39}{8}3^n - 4 \cdot 2^n - \frac{1}{4}n^2 - \frac{5}{2}n + \frac{1}{8}}.$$

## Problem 6

A deposit of \$100,000 is made at the beginning of a year. On the last day of each year two dividends are awarded: 20% of the amount in the account during that year and 45% of the amount in the account in the previous year. Let $P_n$ be the amount at the end of $n$ years.

### (a) Find a recurrence for $\{P_n\}$.

During year $n$ the on-account amount is $P_{n-1}$, so the year-$n$ dividends total $0.20\,P_{n-1} + 0.45\,P_{n-2}$. Hence

$$\boxed{P_n = 1.20\,P_{n-1} + 0.45\,P_{n-2} \quad (n \geq 2), \qquad P_0 = 100{,}000,\ P_1 = 1.20\,P_0 = 120{,}000.}$$

### (b) Closed form for $P_n$.

Solve $P_n - 1.20\, P_{n-1} - 0.45\, P_{n-2} = 0$. The characteristic equation $r^2 - 1.2r - 0.45 = 0$ has roots $r_1 = \frac{3}{2}$ and $r_2 = -\frac{3}{10}$. Therefore $P_n = \alpha \left(\frac{3}{2}\right)^n + \beta \left(-\frac{3}{10}\right)^n$. Using $P_0 = 100{,}000$ and $P_1 = 120{,}000$ yields $\alpha = \frac{250{,}000}{3}$, $\beta = \frac{50{,}000}{3}$. Thus

$$\boxed{P_n = \frac{250{,}000}{3}\left(\frac{3}{2}\right)^n + \frac{50{,}000}{3}\left(-\frac{3}{10}\right)^n \quad \text{dollars.}}$$

## Problem 7

**Claim.** If exactly $n \geq 1$ islanders have blue eyes, then after the foreigner's announcement ("there exists at least one blue-eyed person"), *all and only* the blue-eyed islanders commit ritual suicide, and they all do so simultaneously at noon on day $n$ (counted from the day after the announcement).

**Why the announcement matters.** Before the statement, everyone *sees* blue eyes, but it is not common knowledge that such eyes exist. The announcement makes "there exists a blue-eyed person" common knowledge and thus creates a base case for the inductive reasoning below.

**Proof by induction on $n$.** *Base case $n = 1$.* The unique blue-eyed islander sees no blue-eyed people. After the announcement, the only way the statement can be true is if they themselves are blue-eyed. They therefore commit at noon on day 1. No brown-eyed islander ever gains information that would force them to act. Hence the claim holds for $n = 1$.

*Inductive step.* Assume the claim is true for $1, 2, \ldots, n-1$ (with $n \geq 2$). Suppose there are exactly $n$ blue-eyed islanders. Each blue-eyed islander sees exactly $n - 1$ blue-eyed islanders. Fix any blue-eyed islander $X$ and reason as $X$ does:

- If $X$ were *not* blue-eyed, then there would be exactly $n - 1$ blue-eyed islanders. By the induction hypothesis, all those $n - 1$ would commit at noon on day $n - 1$.
- If noon of day $n - 1$ passes and nobody has committed, that counterfactual is refuted, so $X$ deduces there must be $n$ blue-eyed islanders, i.e., $X$ is blue-eyed.

Because every blue-eyed islander makes the same deduction at the same time, they all commit at noon on day $n$.

Finally, any brown-eyed islander always sees $n$ blue-eyed islanders. They never encounter a refuted counterfactual about "$n-1$ blue-eyed" that would force the conclusion "I am blue-eyed", so they never act. This completes the induction.

**Conclusion for the stated numbers.** Here $n = 100$, so precisely the 100 blue-eyed islanders commit at noon on day 100; the 900 brown-eyed islanders do not. □

## References